

Koszul Rings

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1 Syzygies and Hilbert's Syzygy Theorem

For this discussion, k is a semisimple ring (usually a field), and S is a $\mathbb{Z}_{\geq 0}$ -graded k -algebra with $S_0 = k$. Denote $\mathfrak{m} = S_+ = S_{\geq 1}$ the **irrelevant ideal**. In particular, $S/\mathfrak{m} \cong S_0 = k$ as rings, but this also gives k an S -module structure.

Definition 1.1. A **minimal free resolution** of a graded S -module M is an exact sequence

$$\cdots \rightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \rightarrow 0$$

with P_i free and such that $\text{im} d_i \subseteq \mathfrak{m}P_{i-1}$. In other words, the differentials represented as matrices have only positive degree entries. This is equivalent to the resolution being minimal with respect to chain map inclusion.

We refer to $\text{im} d_i$ as the i th **syzygy module**.

Theorem 1.2. Let $S = k[x_1, \dots, x_n]$ where k is a field. Let M be a finitely generated graded S -module. Then, the n th syzygy module is free. (Equivalently, $P_{n+1} = 0$ in the minimal free resolution).

Proof. Say the minimal free resolution of M is

$$\cdots \rightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} M \rightarrow 0.$$

Since the P_i s are in particular projective, we can use this to compute $\mathrm{Tor}_i^S(M, k)$. Truncating away M and tensoring with $k = S/\mathfrak{m}$ (as an S -module), the differentials vanish and we get

$$\mathrm{Tor}_i^S(M, k) = P_i \otimes_S k,$$

i.e., while $\mathrm{Tor}_i^S(M, k)$ has the structure of a graded S -module, at least as a k -vector space, it is just k^r where r is the free rank of P_i as an S -module (forgetting the grading). So, to show P_{n+1} vanishes, it suffices to show $\mathrm{Tor}_{n+1}^S(M, k) = 0$.

So, we compute $\mathrm{Tor}_{n+1}^S(M, k)$ by a different means, namely resolving k . The sequence $x_1, \dots, x_n$ is a regular sequence, so $\mathrm{Kos}(x_1, \dots, x_n)$ resolves

$$S/(x_1, \dots, x_n) = S/\mathfrak{m} \cong k,$$

at least as an S -module, but in fact we can put gradings on everything, so

$$\mathrm{Tor}_S^{n+1}(M, k) = H_{n+1}(M \otimes_S \mathrm{Kos}(x_1, \dots, x_n)),$$

which is 0 because the Koszul complex has no terms that far. □

2 Duality

We will always denote by $*$ the k -linear dual, with potential sidedness, i.e., V^* for left duals, and $*V$ for right duals (depending on whether the original V was a left or right k -module).

Let X be a graded A - k -bimodule. Define

$${}^{\circledast}X$$

as a k - A -bimodule with graded pieces

$$({}^{\circledast}X)_i = {}^*(X_{-i}),$$

with left k -multiplication by scaling, and right A multiplication by precomposing a functional by left multiplication.

As a set, ${}^{\circledast}X$ is a large subspace of $*X$ that actually carries a grading. (think products vs direct sums issue of finite support)

3 Koszul Ring

First, we give another interpretation of $\mathrm{Kos}(x_1, \dots, x_n)$. In general, we got a Koszul complex on an A -module M by picking a morphism

$$M \xrightarrow{d} A$$

(in our case $S^n \rightarrow S$ by $e_i \rightarrow x_i$) and generating a DGA, i.e., define

$$\wedge^i M \rightarrow \wedge^{i-1} M$$

by Leibniz.

In our case, the variables are symmetric, so we should expect the Koszul complex to at least be S_n equivariant, but even better, it turns out it is GLW equivariant, where $W = \langle x_1, \dots, x_n \rangle_k$. So, our description of this Koszul complex should involve W in a crucial way. To do this, first recognize

$$S = \text{Sym}^\bullet W$$

for $W = \langle x_1, \dots, x_n \rangle$, and let $V = W^* = \langle e_1, \dots, e_n \rangle$ be its dual vector space. Now, a term in the complex is

$$\wedge^i S^{\oplus n} \cong S \otimes_k \wedge^i W \cong S \otimes_k {}^*(\wedge^i V) \cong \text{Hom}_k(\wedge^i V, S).$$

Then, to specify the Koszul boundary, which will be graded $+1$ because all $|x_i| = 1$, we just have to specify the graded pieces. Consider a graded piece as

$$\text{Hom}_k(\bigwedge^j V, \text{Sym}^i W) \rightarrow \text{Hom}_k(\bigwedge^{j-1} V, \text{Sym}^{i+1} W)$$

by the composition

$$\text{Hom}_k(\bigwedge^j V, \text{Sym}^i W) \rightarrow \text{Hom}_k(\bigwedge^j V \otimes W, \text{Sym}^i W \otimes W) \rightarrow \text{Hom}_k(\bigwedge^{j-1} V, \text{Sym}^{i+1} W)$$

where the last map is pullback by contraction $\bigwedge^j V \otimes_k W \rightarrow \bigwedge^{j-1} V$ and multiplication $\text{Sym}^i W \otimes_k W \rightarrow \text{Sym}^{i+1} W$.

(where contraction is first the “differential geometry” embedding

$$\begin{aligned} \bigwedge^j V &\hookrightarrow \bigotimes^j V, \\ \alpha = v_1 \wedge \dots \wedge v_j &\longmapsto \frac{1}{j!} \sum_{\sigma \in S_j} (\text{sgn}(\sigma)) v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(j)} \end{aligned}$$

with image in $\bigwedge^{j-1} V \otimes V$, followed by the evaluation pairing $V \otimes W \rightarrow k$. One can then check that this composition is

$$p \cdot x_{\ell_1} \wedge \dots \wedge x_{\ell_j} \longmapsto p \sum_{i=1}^j (-1)^{i+1} x_{\ell_i} x_{\ell_1} \wedge \dots \wedge \hat{x}_{\ell_i} \wedge \dots \wedge x_{\ell_j},$$

which after unraveling matches the usual Koszul boundary.)

We can then think of the total space of this Koszul complex as

$$\text{Sym}^\bullet W \otimes_k {}^*(\wedge^\bullet V)$$

(for now, because $\wedge^\bullet V$ is finite dimensional, there is no finite support issue, but we do have to worry about the grading/module structure), and the differential is a $(+1, +1)$ graded bimodule morphism (I’m having trouble seeing how this is a right $\wedge^\bullet V$ morphism when k is noncommutative).

4 Internal Hom and Ext Algebras

Let $\mathcal{A}$ be an abelian category (say with enough projectives/injectives), and A an object. Define

$$\mathrm{Ext}^\bullet(A, A) = \bigoplus \mathrm{Ext}^i(A, A)$$

the **Ext Algebra** as an abelian group by this direct sum decomposition (possibly enriched if $\mathcal{A}$ is), and a ring via the **Yoneda cup product**, which can be described in the derived category as follows: given

$$\varphi \in \mathrm{Ext}^i(A, A) \cong \mathrm{Hom}_{D(\mathcal{A})}(A, A[i]), \quad \psi \in \mathrm{Ext}^j(A, A) \cong \mathrm{Hom}_{D(\mathcal{A})}(A, A[j]),$$

define $\varphi \circ \psi$ as

$$\psi[i] \circ \varphi \in \mathrm{Hom}_{D(\mathcal{A})}(A, A[i+j]) \cong \mathrm{Ext}^{i+j}(A, A).$$

This is all nice and good, but how do we compute this?

Let $A^\bullet, B^\bullet \in \mathrm{Kom}(\mathcal{A})$. Define the complex

$$\mathcal{H}om^\bullet(A^\bullet, B^\bullet)$$

by setting (do opposite sign for homology convention?)

$$\mathcal{H}om^i(A^\bullet, B^\bullet) = \bigoplus_k \mathrm{Hom}(A^k, B^{k+i})$$

the collection of degree i maps from graded objects (ignoring the differential), and setting

$$d : \mathcal{H}om^i(A^\bullet, B^\bullet) \rightarrow \mathcal{H}om^{i+1}(A^\bullet, B^\bullet)$$

by $d(f) = f \circ d_A - (-1)^{|f|} d_B \circ f$. In this way, $d(f) = 0$ exactly when $f : A^\bullet \rightarrow B[i]^\bullet$ is a chain map (recall that the differential changes sign per shift). Furthermore, $\mathrm{im}(d) \subseteq \mathcal{H}om^i(A^\bullet, B^\bullet)$ are exactly nullhomotopic chain maps. So,

$$H^i(\mathcal{H}om^\bullet(A^\bullet, B^\bullet)) \cong \mathrm{Hom}_{K(\mathcal{A})}^i(A^\bullet, B^\bullet),$$

and this isomorphism respects composition.

Now, by the lifting properties of projective/injective modules, if we assume either $A^\bullet$ is a complex of projective or $B^\bullet$ is a complex of injectives, then

$$\mathrm{Hom}_{K(\mathcal{A})}^i(A^\bullet, B^\bullet) = \mathrm{Hom}_{D(\mathcal{A})}^i(A^\bullet, B^\bullet).$$

This isomorphism also respects composition, in the sense that the diagram commutes.

$$\begin{array}{ccc} \mathrm{Hom}_K(A^\bullet, B^\bullet) \otimes \mathrm{Hom}_K(B^\bullet, C^\bullet) & \longrightarrow & \mathrm{Hom}_K(A^\bullet, C^\bullet) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_D(A^\bullet, B^\bullet) \otimes \mathrm{Hom}_D(B^\bullet, C^\bullet) & \longrightarrow & \mathrm{Hom}_D(A^\bullet, C^\bullet) \end{array}$$

All this being said, we in some sense can get our hands on the object

$$\mathrm{Ext}_S^\bullet(k, k)$$

by taking a projective resolution $P^\bullet \rightarrow k$, and computing

$$H^\bullet \mathcal{E}nd_S^\bullet(P^\bullet, P^\bullet)$$

as a ring (i.e., multiplication is by taking lifts). (Another way to see that multiplication descends is to observe a Leibniz rule for the differential of a product, to see that nullhomotopic tensors give a nullhomotopic product.)